Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

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Contents

1 Overview **2**

2 Model **2**
- 2.1 State and Firm Types . . . 2
- 2.2 Timing Within Each Period . . . 2
- 2.3 Agglomeration Extension . . . 3
- 2.4 Value Functions and CCPs . . . 3
- 2.5 Solution Method . . . 3

3 Calibration **4**

4 Regional Extension **4**
- 4.1 Regional Structure . . . 4
- 4.2 Solution with Regional States . . . 5

5 Estimation on Simulated Data **6**
- 5.1 Simulated dataset . . . 6
- 5.2 Two-step estimator . . . 6
- 5.3 Results . . . 7

6 Counterfactual: EU–US Alliance **8**
- 6.1 Spillover specification under the alliance . . . 8
- 6.2 Welfare . . . 8
- 6.3 Evaluation by Monte Carlo with common random numbers . . . 8
- 6.4 Calibration and results . . . 9
- 6.5 Discussion . . . 9

7 Next Steps **10**

1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether these effects undo the innovator's dilemma found by Igami.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$.

"Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

   where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect, in that their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{EVT1}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old $\to$ both $\to$ new $\to$ pe):

- old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
- both/new: $\{\text{stay}, \text{exit}\}$
- pe: $\{\text{enter}, \text{quit}\}$

Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t+1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms, capturing the knowledge-spillover or agglomeration channel.

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is effectively a *single-agent problem*: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by backward induction, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\to$ new $\to$ both $\to$ old), exploiting the sequential structure:

- **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. We compute CCPs for all intermediate states reachable from s_0.
- **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
- **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
- **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial / Multinomial draws).

3 Calibration

We summarize the parameters we use for calibration and for simulating the data in Table 1.

Table 1: Parameter values for calibration and simulation

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.3
ϕ	Entry cost	0.2
σ	EVT1 scale	0.5
γ	Agglomeration (default)	0.15
ρ	Cross-market substitution	0.5

4 Regional Extension

4.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm

is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition stays **global**, with all active firms meeting in the same two-market Cournot game of equations (1)–(2). Agglomeration spillovers, in contrast, are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq^2_{oo,r}$, $\pi_{n,r} = bq^2_{nn,r}$, and $\pi_{b,r} = b(q^2_{bo,r} + q^2_{bn,r}) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers. This is a standard **local agglomeration externality**, creating a wedge between the social and private return to innovation and opening the door to regionally targeted industrial policy.

4.2 Solution with Regional States

The sequential-move structure (old $\to$ both $\to$ new $\to$ pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region r decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region R's scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-R continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\mathrm{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\mathrm{EV}_{\text{after new}}$, $\mathrm{EV}_{\text{after both}}$, and $\mathrm{EV}_{\text{after old}}$, each keyed by (State, r). Because each sub-state's CCP is a function of the realized history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6. One visible artefact of the sequential-move convention is a small first-mover asymmetry: at any symmetric s_0 with uniform γ_r, the three regions' CCPs differ by at most 10^{-3}. Region 1 commits without observing anyone, while region 3 best-responds to the realised history. The gap is well below the magnitude of every counterfactual difference reported in Section 6.

5 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate (γ, κ, ϕ), treating the demand parameters, the EVT1 scale σ, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

5.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \text{unif}\{0, 1, 2\}^{4R}$ (rejecting empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, and realised profit. For period 1 we additionally record the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

5.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector γ, identification splits cleanly: γ is pinned by the Cournot block and (κ, ϕ) by the discrete actions, conditional on γ. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

Step 1: NLS for γ. For each market we reconstruct s_0 and s_1 by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\text{slot}} \left(q^{\text{obs}}_{m,t,r,\text{slot}} - \hat{q}_{m,t,r,\text{slot}}(\gamma)\right)^2, \tag{7}$$

starting from $\gamma^0 = (0, 0, 0)$ and leaving γ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 4.88e-14$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

Step 2: MLE for (κ, ϕ). Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over (κ, ϕ). The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP

integrated over the forward traversal). Per market,

$$\begin{aligned}\ell_m(\kappa,\phi) = &\sum_{r=1}^{R}\Big[k_r^{\mathrm{so}}\log p_{\mathrm{so}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{io}}\log p_{\mathrm{io}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{eo}}\log\big(1-p_{\mathrm{so}}-p_{\mathrm{io}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{sb}}\log p_{\mathrm{sb}}(\tilde{s}_r^{\mathrm{BOTH}}) + k_r^{\mathrm{eb}}\log\big(1-p_{\mathrm{sb}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{sn}}\log p_{\mathrm{sn}}(\tilde{s}_r^{\mathrm{NEW}}) + k_r^{\mathrm{en}}\log\big(1-p_{\mathrm{sn}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{ep}}\log p_{\mathrm{ep}}(\tilde{s}_r^{\mathrm{PE}}) + k_r^{\mathrm{op}}\log\big(1-p_{\mathrm{ep}}\big)\Big],\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region r and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating s_0 with the realised outcomes of earlier regions and stages. The total objective is $\ell(\kappa,\phi) = \sum_m \ell_m(\kappa,\phi)$, maximised by Nelder-Mead starting from $(\kappa^0,\phi^0) = (1.0, 1.0)$. The starting point is intentionally far from the calibration, so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

5.3 Results

Table 2 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.1500, 0.1500, 0.1500)$ exactly (Step 1), and the action MLE delivers $\widehat{\kappa} = 0.3002$ (s.e. 0.0416, truth 0.3000) and $\widehat{\phi} = 0.1777$ (s.e. 0.0407, truth 0.2000). Both κ and ϕ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is -1851.916.

Table 2: Two-step estimation of (γ, κ, ϕ)

Parameter	Truth	Estimate	S.E.
κ	0.3000	0.3002	0.0416
ϕ	0.2000	0.1777	0.0407
γ_1	0.1500	0.1500	—
γ_2	0.1500	0.1500	—
γ_3	0.1500	0.1500	—

Notes: Estimates from 500 simulated markets. Step 1 identifies γ from Cournot quantities; Step 2 estimates (κ, ϕ) by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

6 Counterfactual: EU–US Alliance

Pooling regions 1 and 2's innovator counts lifts welfare by roughly +6.44 % in each allied region, leaves region 3 unchanged in welfare terms, and raises aggregate welfare by +4.31 % of baseline. Innovation rates rise by +5.16 % and +4.43 % in the allied regions and fall by $|-4.00|$ % in region 3, so the unallied region is harmed on the real-economy margin even though it breaks even on welfare. The rest of the section sets up the spillover specification, derives the welfare formula, describes the Monte Carlo evaluation, and walks through the mechanism.

6.1 Spillover specification under the alliance

Under the baseline, region r's new-technology marginal cost depends only on its own innovators, equation (6). Under the alliance of $\{1, 2\}$, regions 1 and 2 share an enlarged pool:

$$c^{\text{alliance}}_{n,r,t} = c_{n,0} - \gamma_r \cdot \sum_{r' \in \text{bloc}(r)} \big(n_{b,r',t} + n_{n,r',t}\big), \qquad \text{bloc}(1) = \text{bloc}(2) = \{1, 2\},\ \text{bloc}(3) = \{3\}. \quad (9)$$

The implementation adds a single `blocs` field to the parameter struct; the default `(1,2,3)` recovers equation (6) bit-for-bit.

6.2 Welfare

The two-market linear demand of equations (1)–(2) is rationalised by the quasi-linear utility $U = A(Q_o + Q_n) - \frac{b}{2}(Q_o^2 + Q_n^2) - s\, Q_o\, Q_n$ with $b = B/M$ and $s = \rho B/M$, giving consumer surplus

$$\text{CS}(s) = \tfrac{b}{2}(Q_o^2 + Q_n^2) + s\, Q_o\, Q_n. \quad (10)$$

Because Cournot is global, CS is a single number; we allocate it equally across regions (equal-population assumption). Producer surplus is local: each firm's per-period profit is charged to its own region. Innovation and entry costs are paid by the firm's region in period 1. Two-period discounted welfare per region is

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\, \text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\, \text{PS}_r(s_1) - \kappa\, k_r^{\text{innov}} - \phi\, k_r^{\text{enter}}, \quad (11)$$

which satisfies $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r (\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}})$ by construction. The runner verifies this identity numerically and finds residuals at machine precision.

6.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives both scenarios, and per-market we use `MersenneTwister(seed + k)`. Identical s_0 draws and identical EVT1 shock paths under both scenarios mean each market contributes a paired observation $(W_r^{\text{base}}, W_r^{\text{alli}})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to about 0.2 percentage points by $K = 1000$.

6.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 3 reports the full per-region decomposition. Innovation rises by +5.16 % in region 1 and +4.43 % in region 2, and falls by $|-4.00|$ % in region 3. Entry rates move in the same direction with smaller magnitudes.

Table 3: EU–US alliance: per-region equilibrium and welfare

	Region 1 (US)		Region 2 (EU)		Region 3 (RoW)	
Quantity	Baseline	Alliance	Baseline	Alliance	Baseline	Alliance
$P(\text{innov} \mid \text{old})$	0.3507	0.3688	0.3495	0.3650	0.3533	0.3392
$\Delta P/P_0$	+5.16%		+4.43%		-4.00%	
$P(\text{enter} \mid \text{pe})$	0.5562	0.5723	0.5801	0.5968	0.5792	0.5624
$\Delta P/P_0$	+2.91%		+2.87%		-2.91%	
PS_r	0.8500	0.9290	0.8488	0.9290	0.8800	0.8147
Costs paid	0.0905	0.0942	0.0958	0.0993	0.0958	0.0925
W_r	2.1358	2.2734	2.1293	2.2682	2.1606	2.1608
$\Delta W_r/W_{r,0}$	+6.44%		+6.52%		+0.01%	

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, uniform $\widehat{\gamma}_r = 0.1500$. Baseline blocs $(1, 2, 3)$; alliance blocs $(1, 1, 2)$. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

6.5 Discussion

Three points are worth highlighting.

Allied regions gain in two ways. Their pooled innovator count lowers $c_{n,1}$ and $c_{n,2}$ whenever *either* region has an innovator, raising the period-2 profit of being on new tech. This pulls more firms into innovation and entry there, which raises producer surplus, and the additional new-tech output also lowers prices globally, raising the regions' share of consumer surplus. Welfare rises by +6.44 % in region 1 and +6.52 % in region 2.

Region 3 loses on innovation but breaks even on welfare. Tougher global Cournot competition from cheaper allied competitors drags region 3's $P(\text{innov})$ and $P(\text{enter})$ down by about 4 % and 3 % respectively, and its PS_3 falls by roughly 7 %. Its share of the global CS gain is almost the same magnitude in the opposite direction, so its net welfare moves by

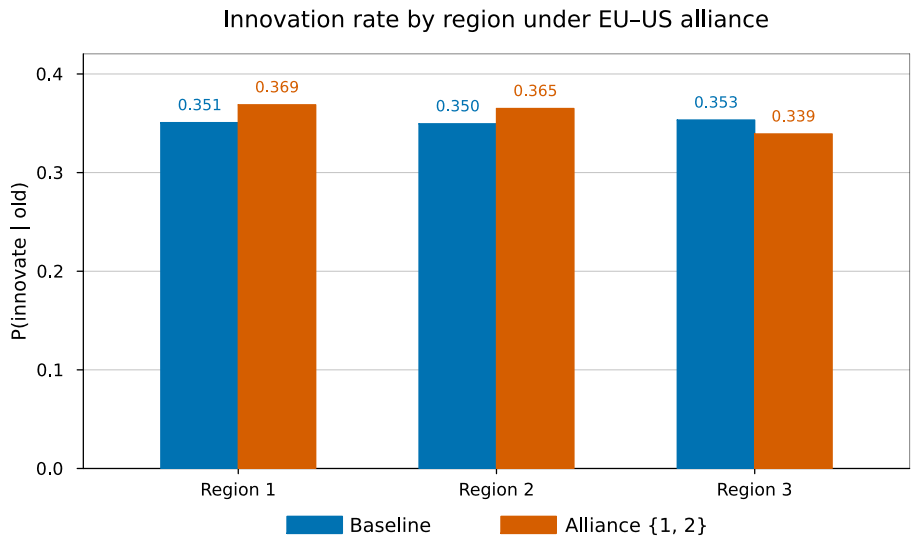

Figure 1: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region, baseline vs. alliance.

less than a basis point (+0.01 %). The unallied region is harmed in real-economy terms (less innovation, lower profits) but the consumer-side benefit of cheaper goods produced by the alliance compensates almost exactly.

The aggregate gain is positive and sizeable. Total welfare rises by +4.31 % of baseline. Roughly half of the gain accrues to each allied region, with a near-zero residual from region 3. Knowledge integration is a positive-sum game even before factoring in the aggregate consumer benefits.

The result depends on the calibration. Under an earlier $(\gamma, \sigma) = (0.05, 1.0)$ setting the same alliance moved welfare by under 1 % everywhere; tripling γ scales the spillover delivered by the merger, and halving σ doubles the logit slope $P(1-P)/\sigma$ against any payoff change, so the combined effect on equilibrium quantities is roughly 6×. The recalibrated values stay well below the cost-floor saturation that sets in at $\gamma \gtrsim 0.30$, so the model is in its responsive interior. Full details, including the K-stability diagnostic and the $\sigma \cdot \gamma_{\text{em}}$ welfare convention, are in `notes/merger_counterfactual.md`.

7 Next Steps

1. **Heterogeneous absorptive capacity.** The alliance result here uses uniform $\widehat{\gamma}_r$. Allowing γ_r to differ across regions, with the bloc-pooled spec of equation (9), would let one region's higher absorptive capacity amplify the alliance's effect.

2. **Fiscal cost of policy.** The current welfare accounting prices in firms' own costs paid but does not net out the fiscal cost of any subsidy a government would need to

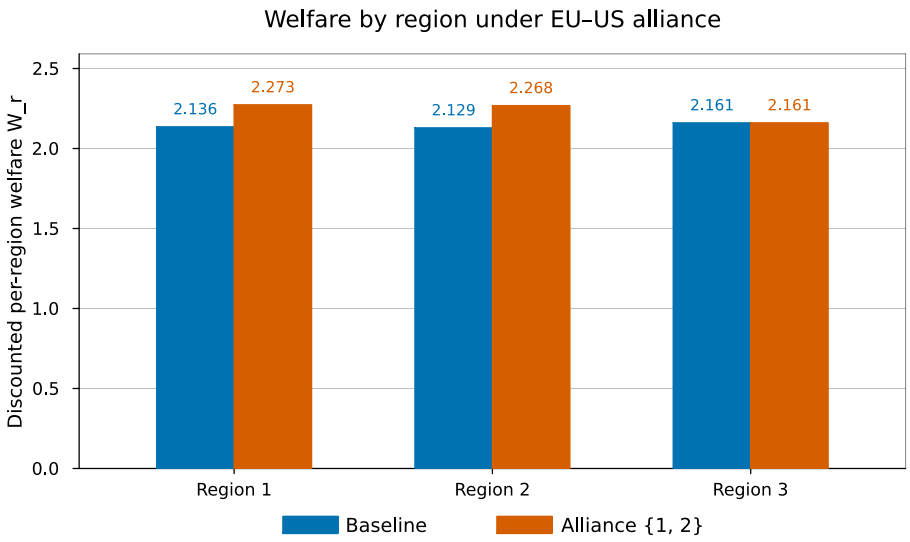

Figure 2: Per-region discounted welfare W_r baseline vs. alliance. Both allied regions gain visibly; region 3's bars are indistinguishable.

deliver to enlarge γ_r in a target region. Adding government revenue and cost makes the targeted-subsidy and alliance scenarios directly comparable as policy levers.

3. $T > 2$ **horizon.** A genuine multi-period extension would force the value-function recursion to use the $\sigma \log \sum \exp$ ex-ante form at every internal period. The current 2-period setup avoids this because period 2 is terminal.